AI-Powered Industrial Data Enrichment

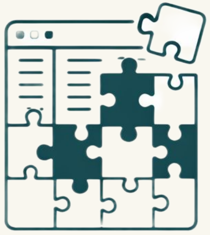
Cut data cleanup from months to hours and build the foundation for smarter business.

Latest update 10.4.2026

data design

AI Advisory &
Implementation Agency

The Silent Cost of Disconnected Industrial Data



Incomplete Data

Technical details like manufacturer, power rating, and pressure are missing or inconsistent.

Manual Search

Teams spend hours hunting for the right device data details and spare parts.

Incomplete BOMs

Quotes and maintenance plans are built on guesswork.

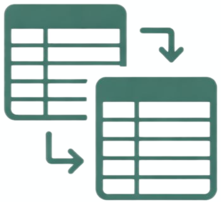
Lost Margin

Mispriced quotes, wrong parts ordered, and delayed maintenance.

AI-Powered Industrial Data Enrichment

With Industrial Data Enrichment, industrial organizations transform incomplete equipment registers into rich, structured asset data – enabling smarter maintenance, better procurement, and data-driven decisions.

Clean Master Data at Scale



10000 device records standardized in hours – a job that used to take a dedicated person months.

Accurate Maintenance Quotes

A service manager uploads a device list and gets complete technical specs in minutes – cutting quote preparation time by 80%.

Spare Part Mapping

A service manager selects a device type and instantly sees the spare parts needed for maintenance – eliminating hours of manual catalog searching per quote.

How it Works?

AI identifies manufacturer, model, device type, segment, dimensions and 20+ fields – from just a device name.

Compair L15-10A K1 ruuvi-kompressori

Parse

AI extracts known patterns from device names and codes

Search

Looks up specs from manufacturer databases and datasheets

Enrich

Fills missing fields, standardizes categories and units

Manufacturer: Ingersoll Rand
Power: 15 kW
Pressure: 10 bar
Voltage: 400 V
Capacity: 2.4 m³/min
Weight: 345 kg
Maintenance interval: 12 m
. . .

Raw device name

All powered by AI – including rule generation

Structured output

Who Needs Structured Asset Data?

Service & Aftermarket Companies

Maintain thousands of devices with incomplete technical data.

Need: Structured, enriched device data for reporting and contract pricing.

Faster quotes, better margins, reliable reporting.

Industrial Distributors

Wide item catalogs from multiple suppliers. Inconsistent device and master data across systems.

Need: Standardized, clean catalogs to operate efficiently.

Clean catalogs, fewer duplicates, efficient procurement.

Production Plants

In-house maintenance organizations with legacy equipment registers full of gaps.

Need: Reliable asset data for maintenance planning.

Proactive maintenance planning and spare-part optimization.

Case Sarlin

Challenge

Sarlin maintained a register of 2500 devices with minimal structured data – device names existed, but manufacturer, power rating, type classification, and other technical details were largely missing or inconsistent. This impacted maintenance contract pricing, spare part planning, and reporting accuracy.

We deployed the Device Data Enrichment tool to

- Enrich each device with 17 new technical fields
- Standardize device types, categories and manufacturers across the entire register
- Provide confidence scores so domain experts could focus review on uncertain results

- **2500 devices enriched in under 3 hours**
- Standardized categories and types aligned with the company's operational taxonomy
- Data quality visibility through fill-rate dashboards and distribution views
- **50% improvement in accuracy in maintenance contract pricing ML model – directly from better data**

Before and After

Today	With the Enrichment Platform
Manual updates by searching spec sheets, websites and product catalogs.	Automatic enrichment of manufacturer, model, power, pressure, voltage and other data from the device name.
Device types and categories inconsistent across the register.	Standardized taxonomy aligned perfectly to your operations.
No visibility into data quality or completeness.	Live data quality dashboards tracking fill rates and confidence scores.
Duplicate manufacturers and inconsistent naming (e.g., "Gardner Denver" vs "Gardener D.")	AI Agents merge and normalize values across the entire register.

Key Features

Customizable enriched fields

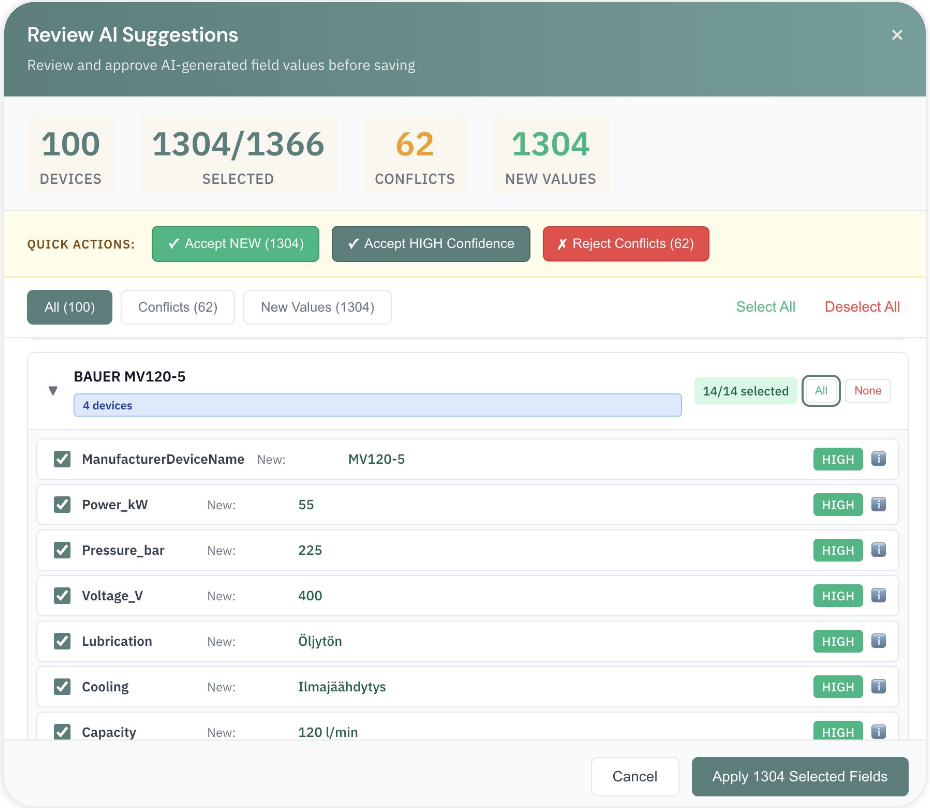
By default covers manufacturer, model, device type, equipment category, industrial segment, dimensions, weight, maintenance interval, and more – fully customizable to client needs.

Enrichment pipeline

Automatically extracts power (kW), pressure (bar), voltage, cooling type, lubrication, and more from device names.

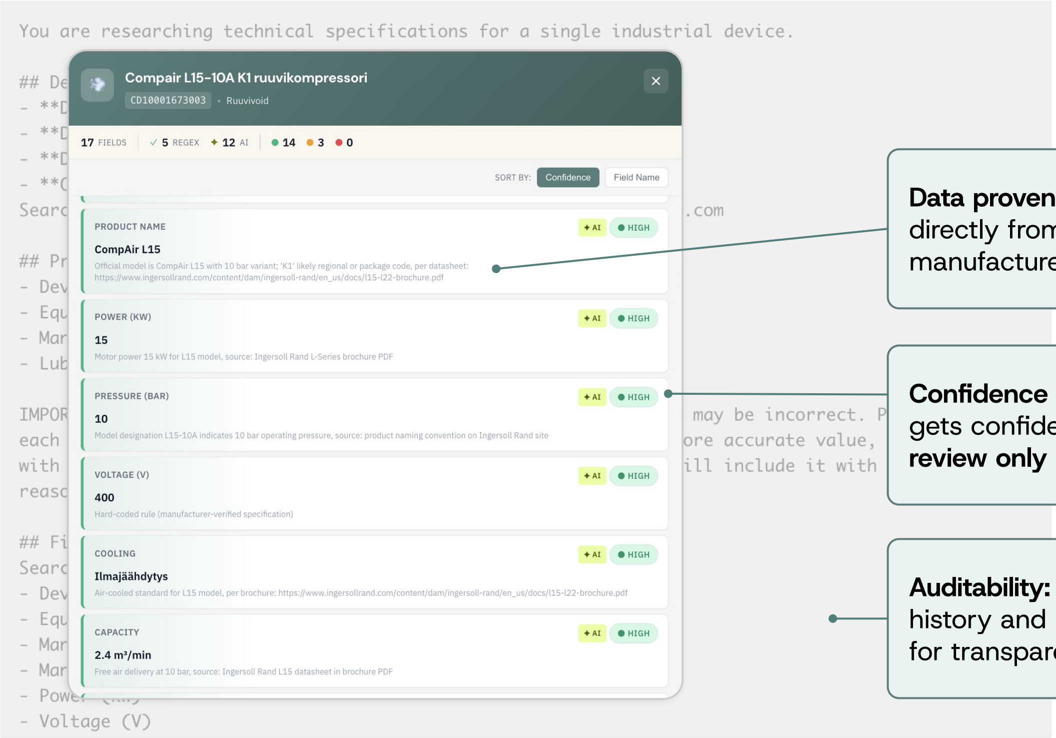
Pre-trained on industrial data

Pre-trained on industrial compressors, air treatment, dryers, and more. Easily extensible to other domains.



Trust & auditability: Human-in-the-Loop

AI suggests, humans approve. No automatic, untraceable changes are pushed to your production systems.



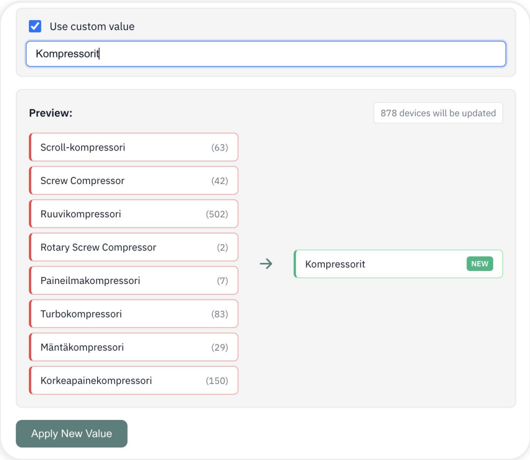
Data provenance: Transparent reasoning directly from the data origin (e.g. linked manufacturer product catalogs).

Confidence scoring: Every enriched field gets confidence score. **Your experts review only what matters.**

Auditability: Complete processing history and LLM reasoning are logged for transparency.

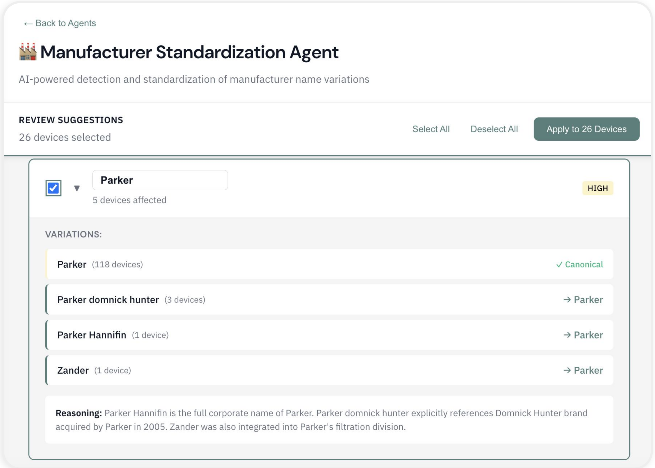
Intelligent consolidation: Built-In AI Agents

Bulk Operations



Manually or automatically consolidate duplicate fields and normalize values across thousands of records instantly.

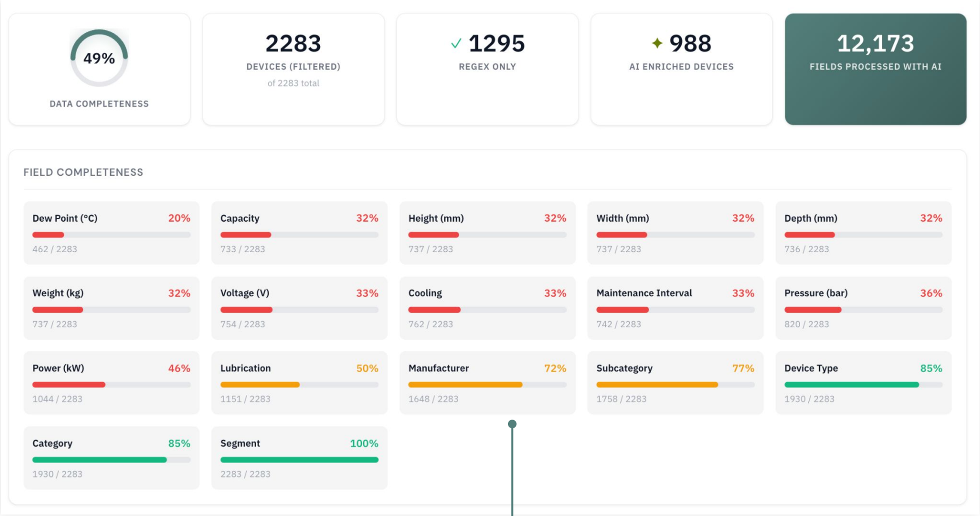
Intelligent Reasoning

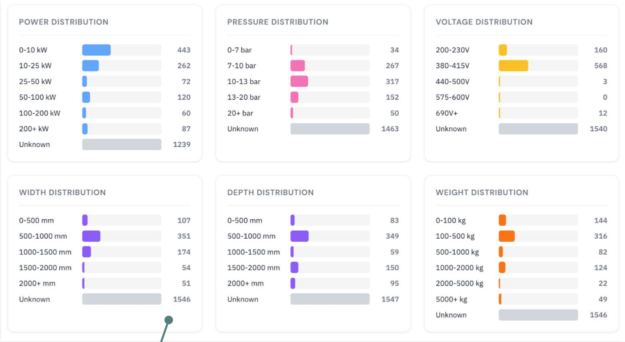


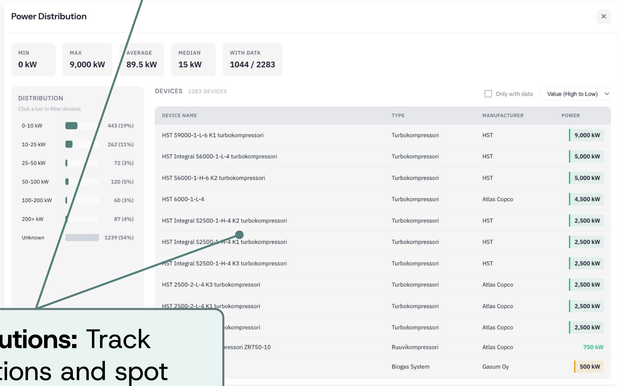
The AI Agents don't just change names, they log the reasoning (e.g. noting that Zander was integrated into Parker's filtration division).

Visibility into your data quality and completion rates

Visualize field fill rates, data distributions to get a complete picture of the data quality.







Fill rates: Instantly see which technical fields are missing across thousands of records.

Distributions: Track distributions and spot anomalies in your fleet.

How It Creates Business Value

Faster Data Cleanup

Transform thousands of records in hours, not months, allowing your team to focus on high-value work instead of data entry.

Higher Data Quality

Standardized categories and confidence scores mean you can finally trust your reports and make decisions based on real data.

Better Maintenance & Procurement

Accurate device data for quotes and spare parts ordering leads to better margins and fewer costly surprises in the field.

Cost-Efficient AI

Targeted AI calls paired with human review. You only pay for value added, not unnecessary processing.

Product roadmap – Coming soon

Spare Part & Item Mapping

Build a cross-reference between internal codes, supplier codes, and manufacturer parts.

Automatically identify required spare parts for each device.

Price Enrichment

Parse supplier price lists into structured data, match via the foundational registry, and refresh device pricing automatically.

Substitute Identification

Suggest alternative parts when prices rise, lead times increase, or availability drops.

Industrial Data Enrichment (The Foundation)

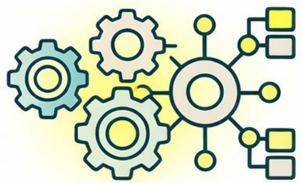
Deployment models

Stand-alone Data Cleanup Tool

Upload equipment registers, enrich and standardize, export clean data.

Best for: Ideal for one-time or periodic master data improvement projects. Immediate triage.

Embedded in Asset Management Workflows



Integrate enrichment directly into your existing data platform. New devices entering the register are automatically enriched, categorized and linked.

Best for: Continuous, automated data governance and foundation for AI use cases.

Enterprise-Grade Security and Control

Your Data, Your Control

Deployment available within your own local environment.

Containerized Isolation

Docker-based deployment ensures clean, secure, and isolated operation from other systems.

Complete Audit Trails

Every AI decision, source link, and confidence score is permanently logged.

Human-in-the-loop

AI acts as an advisor. No automatic data writes to production ERPs without expert approval.

Pilot Partnership

Pilot Partnership

Both sides invest, both sides win. A shared commitment to proving the value with your real data.

You get

Your equipment register enriched with complete technical specifications of your choice and validated against your domain expertise.

A measurable improvement in data quality that directly impacts e.g. maintenance quoting, spare part planning, and reporting.

We get

Real-world industrial data to validate and sharpen our enrichment engine across new device categories and manufacturers.

A reference case with measurable results, and the right to use derived, non-confidential insights to improve the product.

Your investment

3000 – 8000 €

depending on register size

Covers configuration, taxonomy alignment workshop, AI processing costs, and validation support.

Data Design invests separately in product development – that cost is ours, not yours.

Pilot fee is credited toward first 6 months of continued platform use.

After the pilot, you keep all enriched data. No lock-in.

Pilot Roles & Responsibilities

Customer

Test data & golden dataset
Representative 50–100 device sample + 10–20 manually verified records for accuracy measurement

Taxonomy & terminology
Categories, types, naming conventions, abbreviations, and Finnish/English preferences

Domain expert
Named person with allocated 2–4 hours/week for validation rounds and feedback during pilot

Business priorities
Which enriched fields are critical (pricing, spare parts, maintenance planning) and source/target system details

Confidentiality scope
What data can be used for model improvement vs. strictly confidential

Data Design

AI enrichment engine
Functional device data enrichment platform with Parse → Search → Enrich pipeline as foundation

Product development
Dedicated development effort funded by Data Design, including domain extension for customer's device types

Taxonomy alignment
Workshop (1 session) to configure categories, types, and standardization rules to customer's operational model

Accuracy reporting
Field-level confidence scores and accuracy metrics measured against customer's golden dataset

Enriched data & transition
Customer retains all enriched data after pilot. Continued access at discounted rate

Pilot scope: 4–6 weeks • Typical dataset: 50–500 devices • Outcome: validated enrichment + reference case

Get in touch with us!

AI advisory &
implementation agency

data design

+358 40 590 6949
mika.aho@datadesign.fi

datadesign.fi